

Adding Two-Digit Numbers with Place-Value Strategies

Directions:

- Break each number into **tens** and **ones**, add the pieces, then put them back together.
- Fill in every blank — the small blanks show your thinking, and they lead you to the sum.

- 1.** Add $34 + 25$ by breaking both numbers apart.

tens: $30 + 20 =$ _____ ones: $4 + 5 =$ _____

put the pieces together: $34 + 25 =$ _____

- 2.** Add $41 + 37$ by breaking both numbers apart.

tens: $40 + 30 =$ _____ ones: $1 + 7 =$ _____

put the pieces together: $41 + 37 =$ _____

- 3.** Add $26 + 48$. This time the ones make more than a ten!

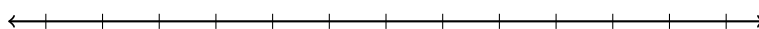
tens: $20 + 40 =$ _____ ones: $6 + 8 =$ _____

put the pieces together: $26 + 48 =$ _____

4. Dara shows $37 + 25$ with base-ten blocks: 3 rods and 7 cubes for the first number, 2 rods and 5 cubes for the second.

- (a) How many *cubes* (ones) are there in all? _____
- (b) Trade ten cubes for one rod. What is the sum? _____

5. Hop along the number line to add $46 + 38$: first one big jump of 30, then a jump of 8.
start at 46, jump +30: land on _____ then jump +8: land on _____



6. Mia adds $39 + 24$ with a friendly number: she adds $40 + 24$ first, then takes away one.
 $40 + 24 =$ _____ so $39 + 24 =$ _____

Explain why Mia takes away one at the end.

7. Use the place-value chart to add $57 + 36$.

	tens	ones
first number	50	7
second number	30	6

tens: $50 + 30 =$ _____ ones: $7 + 6 =$ _____

sum: $57 + 36 =$ _____

8. Challenge. Add three numbers: $45 + 27 + 13$.

$$45 + 27 + 13 = \underline{\hspace{2cm}}$$

Which two numbers did you group first? Explain why that pair made the adding easier.